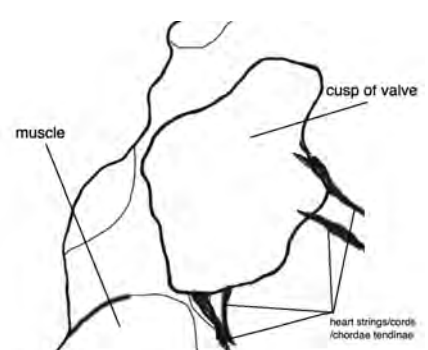


Question Number	Answer	Additional Guidance	Mark
3(a)(i)	• A – right atrium • B - aorta • C - pulmonary artery • D – coronary artery	Any two correct for one mark	(2)

Question Number	Answer	Additional Guidance	Mark
3(a)(ii)	A drawing showing the following features: • cusp and two sets of cords shown (1) • one feature correctly labelled (1) • another feature correctly labelled (1)	valve/muscle/cord/strings/tendon/ventricle/ papillary muscle 	(3)

Question Number	Answer	Additional Guidance	Mark
3(a)(iii)	An answer containing two of the following points: • ventricle / muscle /wall {pumps blood/empties ventricle} (1) • (atrioventricular) valve prevents {blood flowing from ventricle into atrium/backflow into atrium} (1) • cords prevent valve from {opening wrong or closing wrong way/going inside out/flipping} (1) • papillary muscle pulls on cords (1)	allow ecf from diagram, e.g. atrium pumps blood into ventricle	(2)

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	A calculation showing the following steps • correct measurements of lines M from the diagrams and subtraction (1) • calculation of percentage increase (1)	normal, 0.7/0.8, HC, 1.7/1.8 $1.7 - 0.7 = 1.0$ cm/$1.8 - 0.7 = 1.1$/$1.7 - 0.8 = 0.9$/$1.8 - 0.8 = 1.0$ $1.0 \div 0.7 = 140$ $1.1 \div 0.7 = 160$ $0.9 \div 0.8 = 110$ $1.0 \div 0.8 = 130$ (%) ecf 0.6 or 0.9, 1.6 or 1.9 for 1 mark	(2)

Question Number	Answer	Additional Guidance	Mark
3(b)(ii)	An answer that includes the following points; Similarities • right ventricles {unaffected/same} (1) • atria {unaffected/same} (1) Differences • left ventricle {wall/muscle} thicker in the HC heart (1) • left ventricle (chamber) {smaller/shorter} in the HC heart (1)	accept both hearts same size if neither mp1 nor 2 given accept R and/or L atria accept converse accept converse	(4)

Question Number	Answer	Additional Guidance	Mark																				
3(b)(iii)	A table showing the following features - headings, thickness with units and categories (1) - raw data correctly entered (1) - spaces for SDs to be entered (1)	units must not be in cells of table e.g. acceptable range shown in some cells, answer must include just one figure in this range <table><tr><th>group</th><th>mean thickness of Artery A / μm</th><th>SD</th><th>mean thickness of Artery B / μm</th><th>SD</th></tr><tr><td>ECH</td><td>340-345</td><td></td><td>520</td><td></td></tr><tr><td>HC</td><td>455-460</td><td></td><td>585-590</td><td></td></tr><tr><td>control</td><td>460</td><td></td><td>600-610</td><td></td></tr></table>	group	mean thickness of Artery A / μm	SD	mean thickness of Artery B / μm	SD	ECH	340-345		520		HC	455-460		585-590		control	460		600-610		(3)
group	mean thickness of Artery A / μm	SD	mean thickness of Artery B / μm	SD																			
ECH	340-345		520																				
HC	455-460		585-590																				
control	460		600-610																				

Question Number	Answer	Additional Guidance	Mark
3(b)(iv)	An answer that includes the following: • suitable manipulation of the data Plus three from the following points • ECH has an {effect / greater effect} on the arteries, HC {does not / has smaller effect} (1) • correct description of effect of ECH on artery A (1) • correct description of effect of ECH on artery B (1) • correct description of effect of HC on arteries A or B (1)	accept ECH has greatest / most effect e.g.ECH causes decrease (in thickness) in artery A e.g ECH has no effect on Artery B / decrease (in thickness) in artery B is not significant e.g HC does not affect artery A thickness / decrease (in thickness) in artery A is not significant HC does not affect artery B thickness / decrease (in thickness) in artery B is not significant	(4)